Online Pentesting Platforms

Your "Hello World" program to the pentesting myriad

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Being a techie youngling, you decided to learn some programming and as you kept mastering one language after another, you noticed that every tutorial starts with a hello_world.cpp, helloWorld.java and all the combinations of other extensions. But man oh man, where's the h3ll0W0rld.leet to begin your journey in the pentesting and cyber-security world?!

The fire built within gets extinguished real quick when no proper beginner-friendly entry points are present in a field you are interested in. According to ISC2, the unfulfilled job positions in the security sector are soaring with a number of 2.8million jobs without suitable applicants in 2018, projected to hit even higher numbers by 2021 . To combat these woeful numbers, we will introduce you to great platforms to push the progress forward in your journey towards becoming a hacker.

A lot of platforms are available on the Internet, we chose only the ones that are the most helpful from a practical viewpoint and are respected in the security community. Mainly, platforms provide facilities to hone your security skills in two major ways: challenges and labs/machines. Challenges are isolated problems focussed on specific topics, for example, to teach buffer overflow techniques, binary file alongside its source code might be provided. These can be looked at as the questions posed in a CTF. On the other hand, platforms providing labs/machine put you in a real-life like environment that mimics the conditions that might be faced by an actual security professional. These labs may be remote or available on a local system. We'll be using machines/labs/boxes interchangeably, so don't get confused as they all mean the same thing.

PLATFORMS PROVIDING LABS

Now let's take a look at some of the platforms which will force you to think like a professional and let you get a taste of how things work in the wild.

1. HackTheBox

[https://www.hackthebox.eu/]

This had to be the first in our list because well, of course, they provide some of the best resources and are highly respected in the security community. HTB really puts your skills to test, even the sign-up procedure requires a bit of hacking! You need an invite code, but unlike every other invite-system, here the invite code is given by the website itself and not by another user, albeit a little reading between the lines is required.

When you succeed in logging in, you will see yourself presented with a dashboard. All the machines are located on a virtual network managed by the platform. So, as to connect their network and access the machines, the dashboard which will provide you the ".ovpn" file, which is used with the OpenVPN tool. This will link your computer through a virtual link to their network. You can check whether you are connected or not by running "ip a" command from a Linux terminal. If you see a "tun0" entry, that means you are in. The IP starting with "10.10.x.x" will be used by the other machines on the network. This will be extremely useful when doing basic exploitation, file transfer, etc.

After you have done your IP configurations, the rest is simple: select a machine(called a box), submit the user and root flag hidden in a box and you will "Own" it. But here's the catch, root flags can be read by users having, as the name suggests, root privileges and user flags can be read by the specific "fictional" users of the box. And to even get to these flags, let alone reading them, you need to exploit, leverage and pivot all the way. At a specific moment, more than one member can be working on a single box. Even though there are a lot of servers available, due to the huge popularity of HTB each server is buzzing with budding hackers, so it's not unseen that someone mistakenly (or deliberately, in case of a jerk hacker!) alter the machine causing hiccups in others' progress. To tackle this, a reset button is provided for each machine that'll set the machine to its default state. These boxes are created by community members and curated by the HackTheBox staff. Boxes' difficulties include levels Easy, Medium, Hard and Insane but these ratings are often misleading where medium boxes are harder than hard ones and insane boxes are easier than the medium ones.